

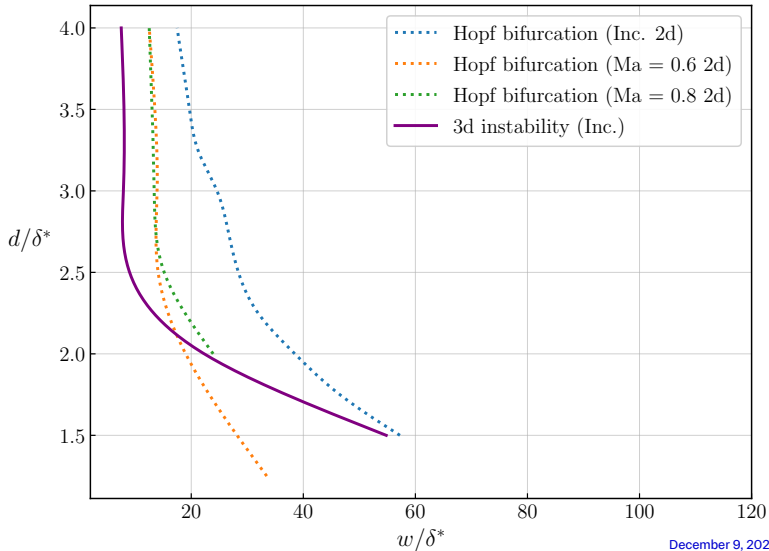
IMPERIAL

jeff update

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Linear instability results

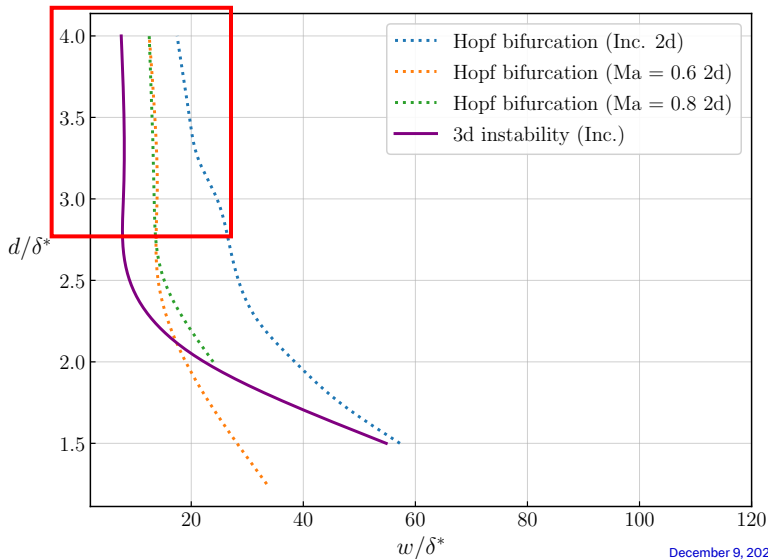
- We know now that the first linear instability to appear in 2d is a Hopf.
- The δ^* depends on Ma . Should we leave it as is or fix it to a standard value (e.g. incompressible value, for example)?



Linear instability results

About limit $d/w \approx \text{large}$

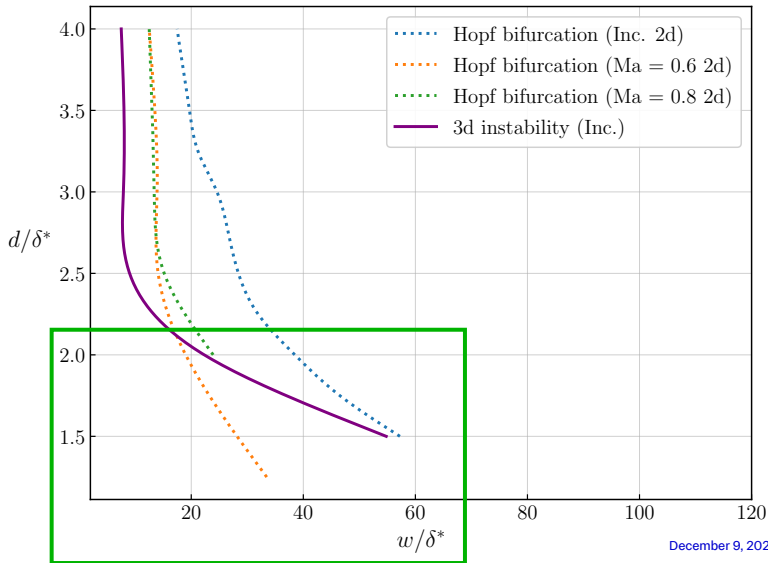
- In general we observe a shift in the onset of instability to lower values of w (at fixed d) when increasing Ma until eventually (I think) reaching a limit (which could be imposed by domain restrictions).
- A 3d instability located in the gap appears before the $Ma = 0.8$ 2d instability.



Linear instability results

About limit $d/w \approx \text{small}$

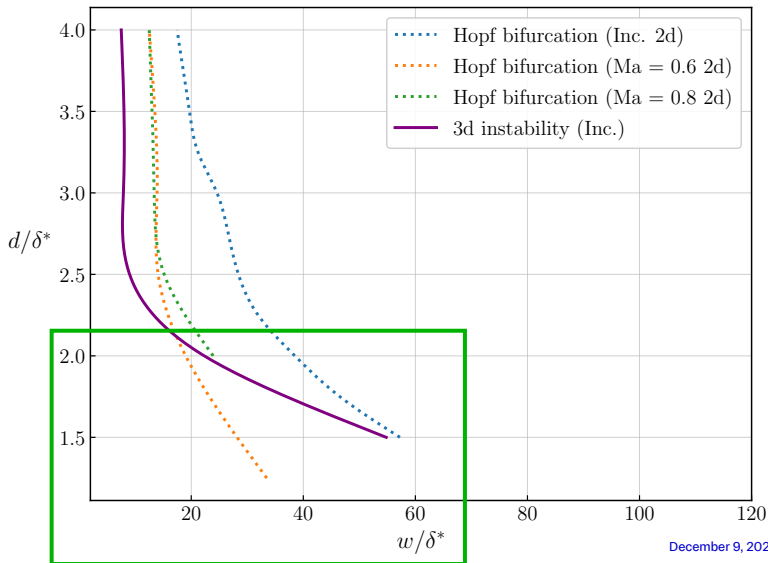
- My feeling is that in the limit, 2d instabilities appear before 3d ones (even in the inc case). In the picture it cannot be properly appreciated because I didn't have enough baseflows to run the 3d stability analysis at higher w values, but all the $d = 1.5$ run in 3d became stable.
- Interestingly, $Ma = 0.8$ bifurcates after $Ma = 0.6$ at some particular depths.



Linear instability results

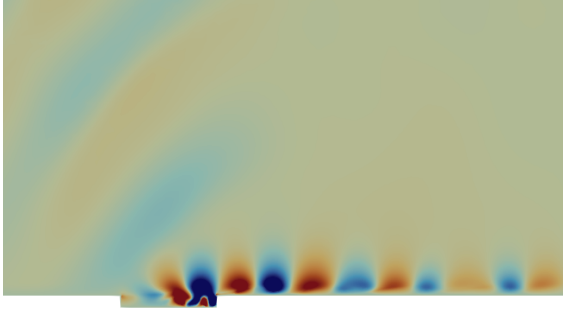
About limit $d/w \approx \text{small}$

- Also instabilities seems to appear at lower d in the compressible case compared to the incompressible runs. For example, there is an unstable configuration at $d = 1.25$ and $w \gtrsim 35$ for $\text{Ma} = 0.6$, while in the incompressible case all the runs at $d = 1.25$ were stable (in the range $w \in [10, 100]$).

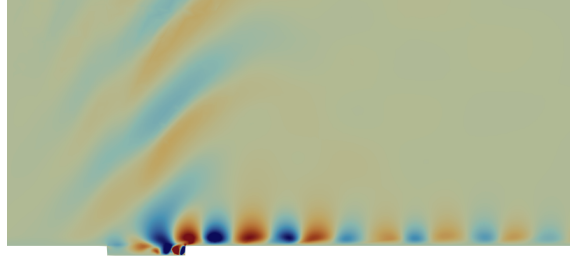


Unstable mode comparison $d = 2.5$

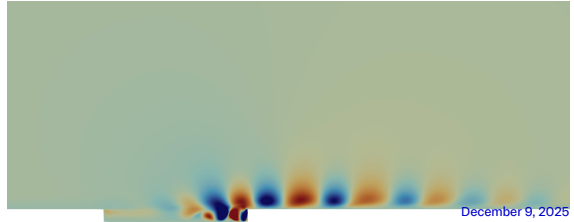
$w = 20$, $Ma = 0.6$:



$w = 20$, $Ma = 0.8$:



$w = 29$, incompressible:



Unstable mode 3d $d = 3, w = 10$

